

1. Student Details

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About Me:

I am a Computer Science Engineering student (dual degree) with a strong interest in web development and problem-solving. I enjoy building full-stack applications and improving user experience through clean UI and efficient backend logic. My goal is to become a skilled developer and work on scalable real-world systems.

2. Project Details

Project Title: Placement Portal Application

Problem Statement:

To design and develop **Placement Portal Application** web application that allows **Admin (Institute), Company, and students** to interact with the system based on their roles.

Approach:

The application was developed using Flask as the backend framework with SQLAlchemy for database management. The system is divided into three main modules:

Student Module: Register, login, apply for jobs, manage profile

Company Module: Register, login, post jobs, manage applications

Admin Module: Approve companies/jobs, view system data

The application follows a modular structure with separate routes, templates, and database models. Jinja2 is used for dynamic HTML rendering.

3. AI/LLM Declaration

I used **ChatGPT (GPT-5.3)** to assist in:

Basic Flask routes definitions, generating boilerplate HTML and Bootstrap templates and took help for complex errors.

The extent of AI/LLM usage is around **10–15%**, limited to **code templates and syntaxes, style suggestions and documentation formatting**.

All final implementation logic, flask routes , debugging, database schemas and integration were done manually.

4. Technologies and Frameworks Used

Technology/Library	Purpose
Flask	Backend web framework
SQLAlchemy	Object Relational Mapper for SQLite database
Jinja2	Template engine for rendering dynamic HTML pages
Bootstrap / CSS	UI styling (Frontend)
SQLite	Database for storing the data
Werkzeug	File upload handling
HTML/CSS	Frontend

5. Database Schema / ER Description

Tables:

1. **Student:** stores student details (id, name, email, password, education, skills, resume)
2. **Company:** stores company profile details (id, name, email, password, website, hr_contact, approval_status)
3. **JobPosition:** stores the created drive details (id, job_title, description, salary, eligibility, status, company_id)
4. **Application:** stores student applications for a particular drive(JobPosition) (id, student_id, job_id, status, application_date)
5. **Placement :** stores the detailed of places students and drive details (id, student_id, company_id, job_id, salary)

Relationships:

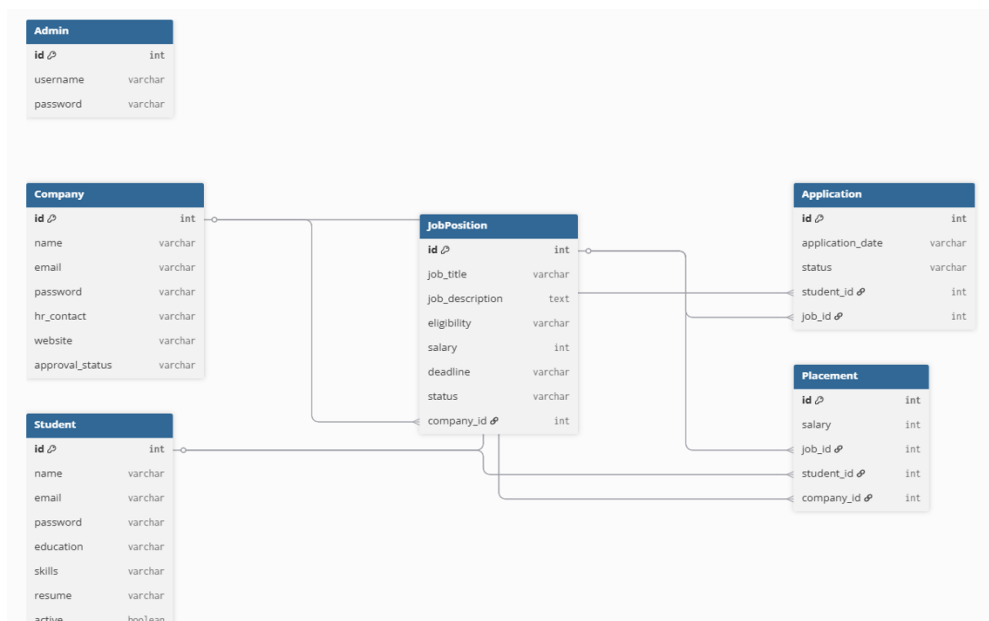
One-to-Many → Company → JobPosition

One-to-Many → Student → Application

One-to-Many → JobPosition → Application

One-to-Many → Student → Placement (when selected)

One-to-One → Application → Placement (later)



6. API Resource Endpoints

Endpoint	Method	Description
/api/students	GET	Get all students
/api/students/<int:id>	GET	Get student by id
/api/companies	GET	Get companies
/api/jobs	GET	Get all jobs
/api/jobs/<int:job_id>	GET	Get job details by id

7. Architecture and Features

Architecture Overview:

- app.py → main Flask app
- models.py → database models
- templates/ → HTML pages and jinja2 dynamic data
- static/ → resume uploads and css(later)
- instance/placement.db → database SQLite

Implemented Features:

- Student registration & login
- Company registration & approval system
- Admin dashboard and control
- Job posting and management
- Apply for jobs
- Application status tracking
- Placement generation after selection
- Resume upload functionality
- Search functionality
- API endpoints
- Avoiding duplicate registrations for both company and student

Additional Features:

- Role-based login system

8. Video Presentation

Drive Link:

https://drive.google.com/file/d/1xbQJuUaQSVFE9CPr_AoKnWdf2EFN2Eag/view?usp=sharing